

Internet of Things (IoT)

Lecture 4: Communication Technologies in IoT

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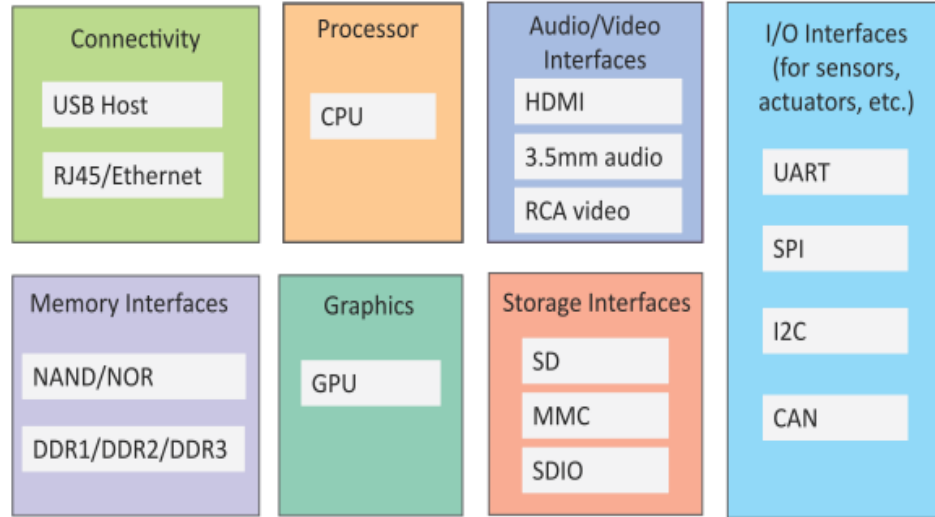
IoT Course Topics

1. Introduction to IoT
2. IoT Architecture & Network Components
3. Smart Objects & IoT Devices
4. **Communication Technologies in IoT**
5. IoT Protocol Stacks
6. IoT Application Protocols & Data Formats
7. IoT Security Principles
8. Cloud & Fog Computing in IoT
9. Data Analytics in IoT
10. IoT Platforms & Middleware
11. Designing IoT Solutions



Generic Block Diagram of an IoT Device

- An IoT device may consist of several interfaces for connections to other devices, both wired and wireless.
- I/O interfaces for sensors
- Interfaces for internet connectivity
- Memory and storage interfaces
- Audio/video interfaces

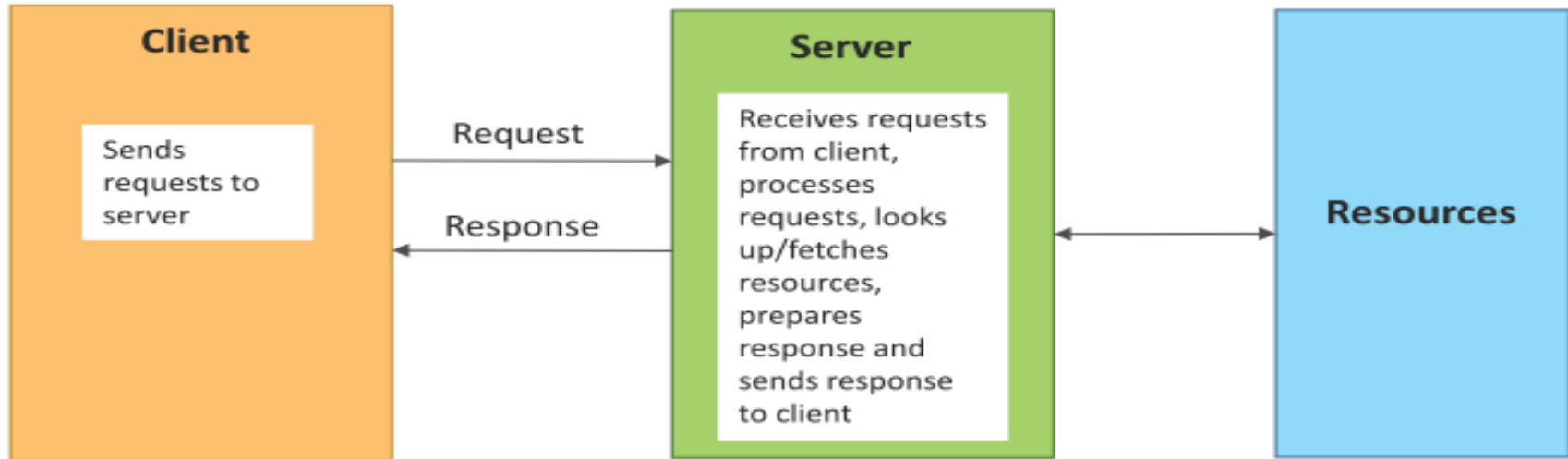


Communication Model

- There are different types of communication model which are:-
 - Request–Response Communication Model
 - Publish–Subscribe Communication Model
 - Push–Pull Communication Model
 - Exclusive Pair Communication Model
 - REST-based Communication APIs

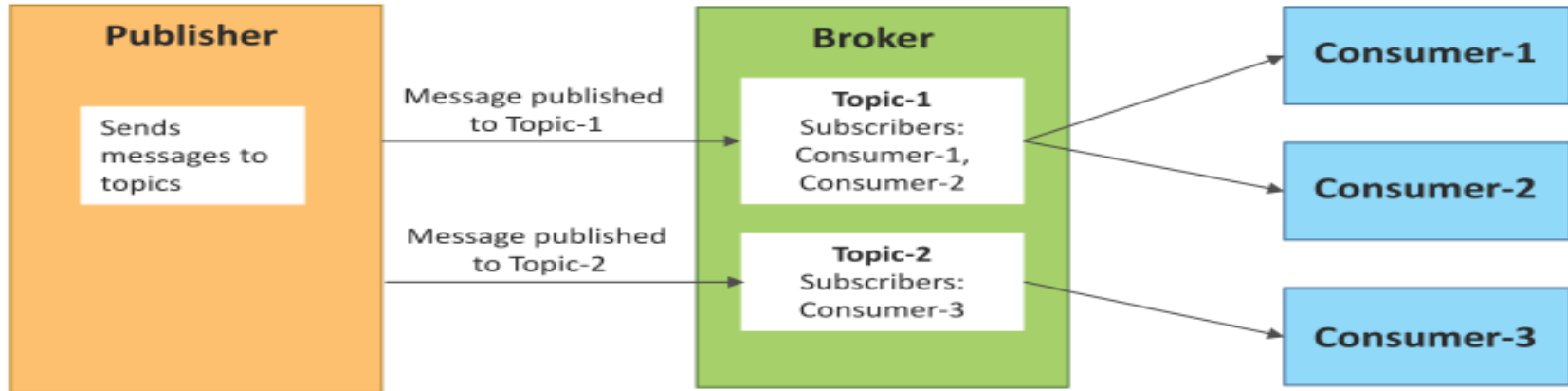
Request–Response Communication Model

- Request–Response is a communication model in which the client sends requests to the server and the server responds to the requests.
- When the server receives a request, it decides how to respond, fetches the data, retrieves resource representations, prepares the response and then sends the response to the client.



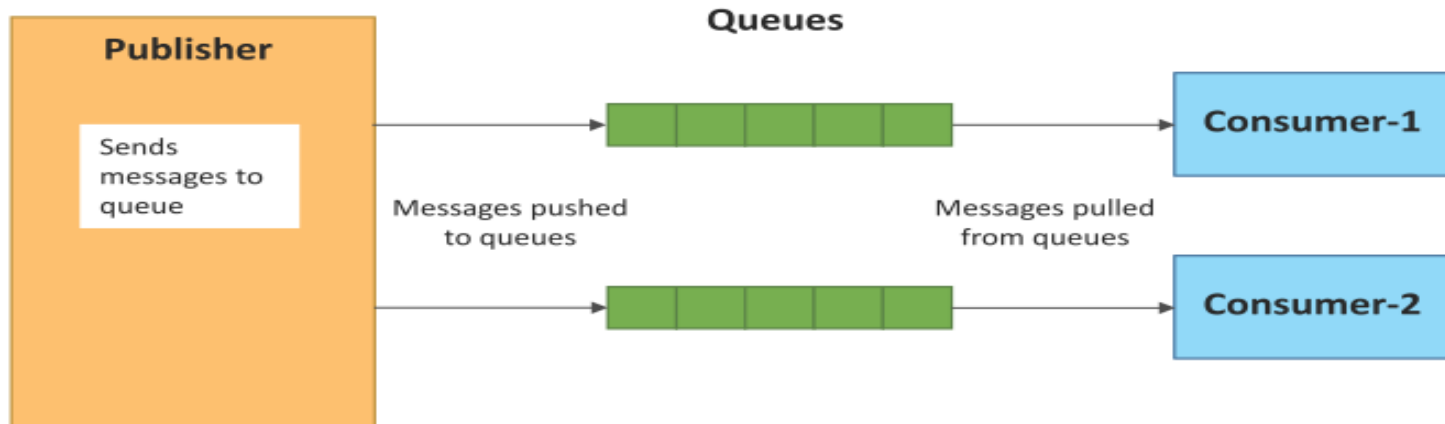
Publish–Subscribe Communication Model

- Publish–Subscribe is a communication model that involves publishers, brokers and consumers (subscribers).
- Publishers are the source of data. Publishers send the data to the topics which are managed by the broker. Publishers are not aware of the consumers.
- Consumers subscribe to the topics which are managed by the broker.
- When the broker receives data for a topic from the publisher, it sends the data to all the subscribed consumers.



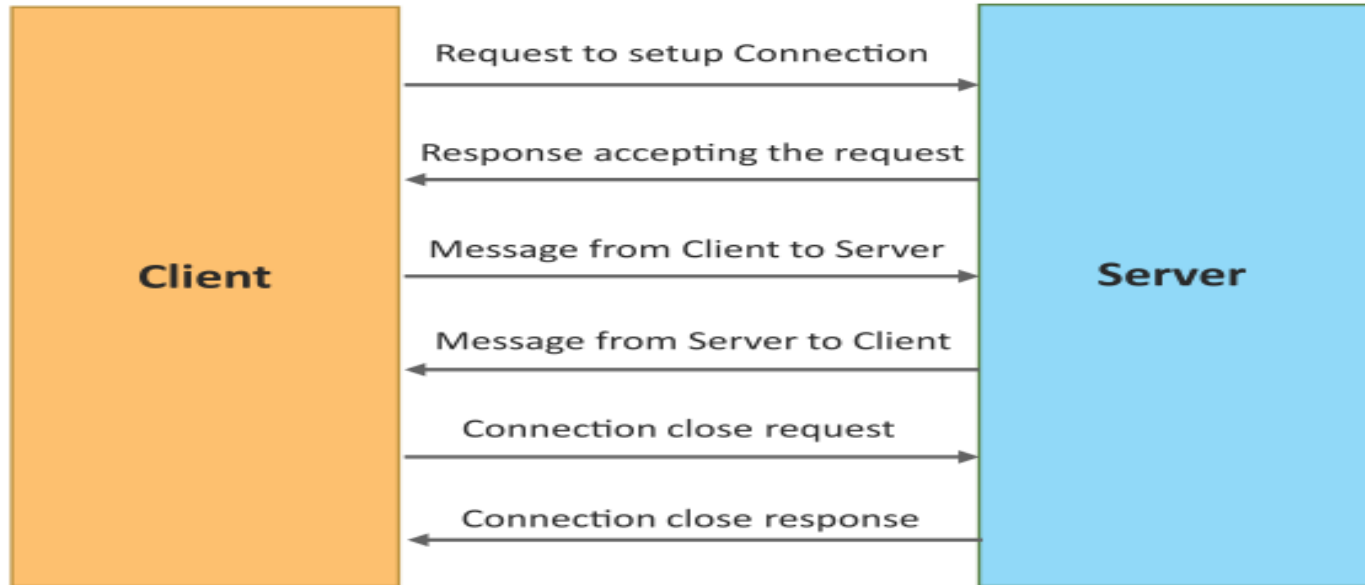
Push-Pull Communication Model

- **Push-Pull** is a communication model in which the data producers push the data to queues and the consumers pull the data from the queues. Producers do not need to be aware of the consumers.
- Queues help in decoupling the messaging between the producers and consumers.
- Queues also act as a buffer which helps in situations when there is a mismatch between the rate at which the producers push data and the rate at which the consumers pull data.



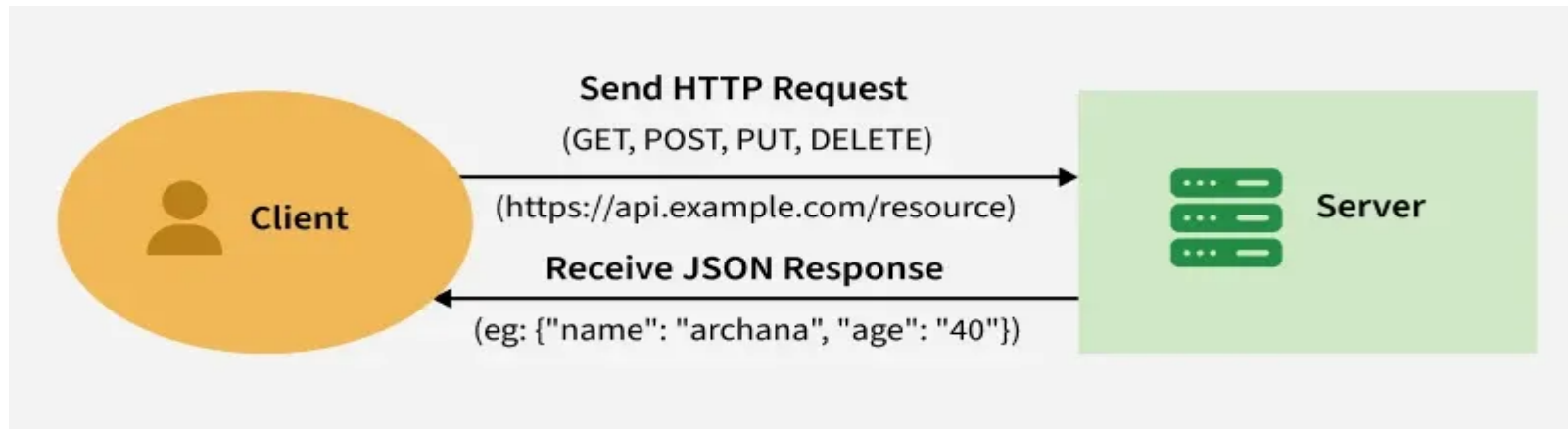
Exclusive Pair Communication Model

- Exclusive Pair is a bidirectional, fully duplex communication model that uses a persistent connection between the client and the server.
- Once the connection is set up it, remains open until the client sends a request to close the connection.
- Client and server can send messages to each other after connection setup.



REST-based Communication APIs

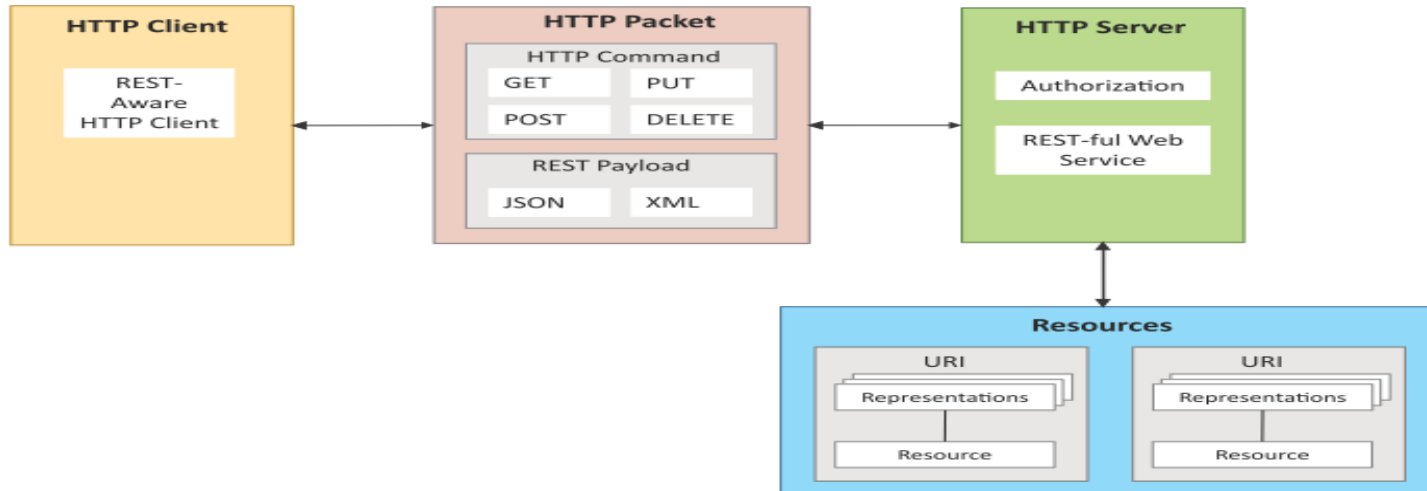
REST API stands for Representational State Transfer API. It is a type of API (Application Programming Interface) that allows communication between different systems over the internet. REST APIs work by sending requests and receiving responses, typically in JSON format, between the client and server.



- A request is sent from the client to the server via a web URL, using one of the HTTP methods.
- The server then responds with the requested resource, which could be HTML, XML, Image, or JSON, with JSON being the most commonly used format for modern web services.
- These methods map to CRUD operations (Create, Read, Update, Delete) for managing resources on the web.

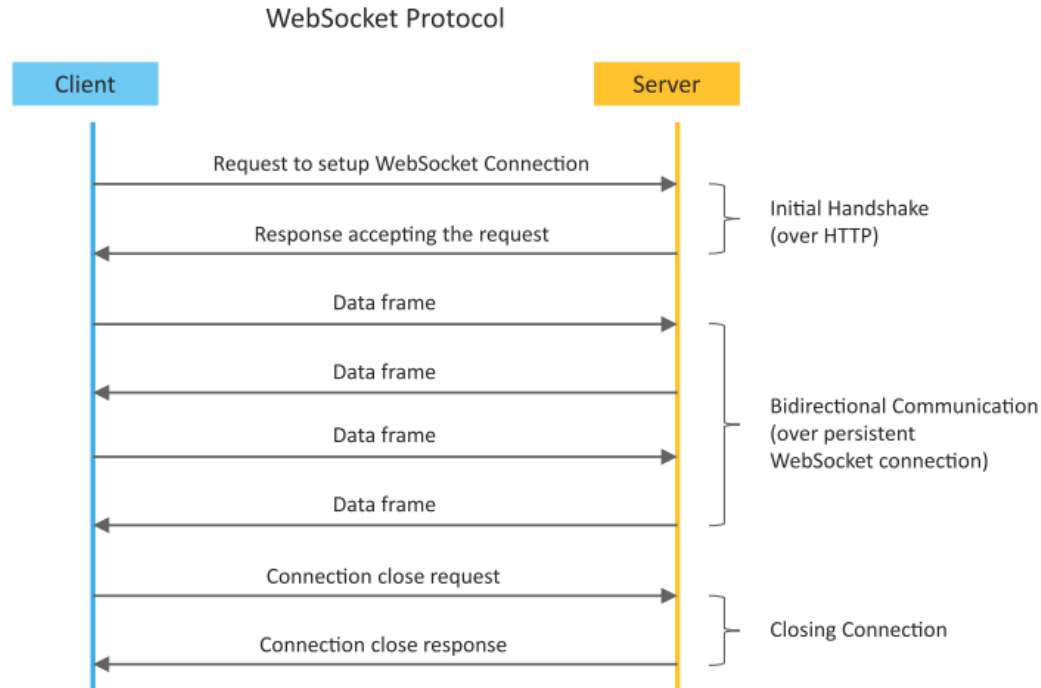
REST-based Communication APIs

- Representational State Transfer (REST) is a set of architectural principles by which you can design web services and web APIs that focus on a system's resources and how resource states are addressed and transferred.
- HTTP is the communication protocol used for data transfer over the web. REST APIs use HTTP methods to interact with resources, but they are not the same thing. REST defines how the APIs should behave, while HTTP defines the rules for communication over the web. They commonly work together, but they serve different purposes.
- REST APIs follow the request-response communication model.



WebSocket-based Communication APIs

- **WebSocket APIs allow bi-directional, full duplex communication between clients and servers.**
- **WebSocket APIs follow the exclusive pair communication model.**





Question
&
Answer

